

School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Lecture-7 Scribe Submission

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CSE400 – Lecture 7 Scribe

Lecture: Expectation, CDFs, PDFs and Problem Solving

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1. Lecture Structure

According to the outline on page 2, the lecture covers:

- Cumulative Distribution Function (CDF)
 - Definition
 - Properties
 - Example
- Probability Density Function (PDF)
 - Definition
 - Properties
 - Example
- Expectation of Random Variables
 - Definition
 - Expectation of function of RV
 - Linearity
 - Moments & Central Moments
 - Variance
 - Skewness
 - Kurtosis
- L7_S2_A_Revised

2. Cumulative Distribution Function (CDF)

2.1 Conceptual Illustration

The diagram on pages 3–4 shows a cylindrical tank analogy:

- Height corresponds to value of random variable
- Accumulated volume corresponds to cumulative probability
- Maximum filled volume corresponds to probability 1

This visual introduces cumulative accumulation behavior.

2.2 Definition

For random variable X :

$$F_X(x) = Pr(X \leq x)$$

This function characterizes the probability behavior of the experiment.

2.3 Properties

Bounds

$$0 \leq F_X(x) \leq 1$$

Limits

$$F_X(-\infty) = 0$$

$$F_X(\infty) = 1$$

Monotonicity

$$x_1 < x_2 \Rightarrow F_X(x_1) \leq F_X(x_2)$$

Interval Probability

$$Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$$

2.4 Example (Validity)

Slides present candidate functions and verify whether they satisfy CDF properties by checking:

- Range constraint
- Monotonic increase
- Proper limits

Functions violating these conditions are rejected.

2.5 Example (Probability Computation)

Given

$$F_X(x) = (1 - e^{-x})u(x)$$

Probabilities computed using:

$$Pr(X > a) = 1 - F_X(a)$$

$$Pr(a < X < b) = F_X(b) - F_X(a)$$

$$Pr(X > a \mid X < b) = \frac{F_X(b) - F_X(a)}{F_X(b)}$$

3. Probability Density Function (PDF)

3.1 Definition

$$f_X(x) = \lim_{\epsilon \rightarrow 0} \frac{Pr(x \leq X < x + \epsilon)}{\epsilon}$$

Using relation with CDF:

$$f_X(x) = \frac{d}{dx} F_X(x)$$

3.2 Fundamental Relationship

$$F_X(x) = \int_{-\infty}^x f_X(y) dy$$

Thus:

PDF = derivative of CDF

CDF = integral of PDF

3.3 Properties

Non-negativity

$$f_X(x) \geq 0$$

Normalization

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

Interval Probability

$$Pr(a < X \leq b) = \int_a^b f_X(x) dx$$

3.4 Examples

Slides show checking candidate PDFs by verifying:

- Non-negative over domain
- Integrates to 1

Conversion examples:

From CDF $\rightarrow$ PDF

Differentiate given CDF expression.

From PDF $\rightarrow$ CDF

Integrate given PDF expression.

4. Expectation of Random Variables

4.1 Definition

Discrete case:

$$E[X] = \sum x_k p_X(x_k)$$

Continuous case:

$$E[X] = \int x f_X(x) dx$$

L7_S2_A.Revised

4.2 Expectation of Function of RV

$$E[g(X)] = \sum g(x_k) p_X(x_k)$$

or

$$\int g(x) f_X(x) dx$$

4.3 Linearity of Expectation

$$E[aX + b] = aE[X] + b$$

5. Moments and Central Moments

5.1 n-th Moment

$$E[X^n]$$

5.2 Central Moment

$$E[(X - \mu)^n]$$

with

$$\mu = E[X]$$

5.3 Variance

$$\sigma^2 = E[X^2] - \mu^2$$

5.4 Higher-Order Shape Measures

Slides identify:

- Skewness
- Kurtosis

as derived from higher central moments.

6. Logical Flow Summary

Lecture progression:

- Introduce cumulative probability representation
- Define and analyze CDF
- Derive PDF and relationships
- Apply through examples
- Introduce expectation and linearity
- Extend to moments and distribution shape